

PD Scorecard v1.0 -- Retail Credit Risk

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Model owner:	Credit Risk Function	Status:	Development / Proof of Concept
Review cycle:	Annual	Regulatory framework:	Basel III IRB

1. Purpose and Scope

This memorandum documents the methodology, assumptions, performance characteristics, and known limitations of a Probability of Default (PD) scorecard developed for retail credit risk assessment. The model is intended as a proof-of-concept implementation of the Internal Ratings-Based (IRB) approach under Basel III, estimating the 12-month PD for individual loan applicants. It is not approved for regulatory capital calculation and has not undergone independent model validation.

The scope covers the PD component of the Expected Loss framework ($EL = PD \times LGD \times EAD$). LGD and EAD are outside the scope of this model.

2. Regulatory Context

Under the Basel III IRB approach (BCBS, 2017), institutions using internal models for credit risk must produce validated PD estimates for each rating grade. The European Banking Authority guidelines on internal models (EBA/GL/2017/16) and the ECB Guide to Internal Models (2019) impose additional requirements on model governance, documentation, and validation. Key regulatory requirements addressed by this implementation:

Explainability: GDPR Article 22 requires that automated credit decisions be explainable to the applicant. The model provides SHAP-based factor explanations for each decision via the /explain endpoint.

Calibration: PD estimates must reflect long-run average default rates (through-the-cycle calibration). Logistic regression outputs are calibrated probabilities by construction.

Auditability: SHAP values for linear models are mathematically exact (not approximate), producing a complete and reproducible audit trail for each scoring decision.

Model validation: EBA guidelines require independent validation including discriminatory power testing, calibration assessment, and stability monitoring. These are partially implemented and identified as next steps.

3. Training Data

The model was developed on the German Credit Data (Statlog) dataset, sourced from the UCI Machine Learning Repository (Hofmann, 1994). The dataset contains 1,000 loan applications from a German bank with 20 applicant and loan characteristics. The binary target variable indicates whether the applicant was classified as a good payer (1) or defaulter (0).

Characteristic	Detail
Observations	1,000
Features (raw)	20
Features (engineered)	5 additional derived features
Target variable	credit_risk (1 = good, 0 = default)
Class distribution	70% good payers / 30% defaulters
Observation period	1990s, Germany
Bureau data	Not available
Macroeconomic features	Not available

Limitation: the dataset is small by production standards and geographically and temporally restricted. A production IRB model would require a minimum observation period of five years of through-the-cycle data under EBA/GL/2017/16. Migration to the Home Credit Default Risk dataset (307,511 applications) is planned.

4. Modelling Methodology

4.1 Feature Engineering

Feature	Construction	Rationale
dti_ratio	Loan amount / income proxy	Debt serviceability indicator
credit_history_length	(Age - 18) + 0.25 x duration	Proxy for length of credit file
has_delinquency	Binary flag from credit history text	Past delinquency indicator
employment_length	Ordinal encoding of employment duration	Income stability proxy
amount_x_duration	Amount x duration / 10 ⁶	Interaction: large long-term exposures

4.2 Model Selection

Four classifiers were evaluated using 5-fold stratified cross-validation: Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Logistic Regression was selected for deployment on three grounds: (i) outputs are calibrated probabilities by construction, satisfying the IRB requirement for PD estimates to reflect observed default rates; (ii) SHAP values are mathematically exact for linear models, producing auditable explanations; (iii) regulatory guidance (EBA/GL/2020/06) expresses a preference for interpretable models in automated credit decisions.

4.3 Class Imbalance

The training data exhibits a 70/30 class imbalance (good:default). Balanced class weights were applied during training (weight inversely proportional to class frequency), which is equivalent to oversampling the minority class and is the recommended approach for credit scoring applications where the cost of false negatives (approving defaulters) exceeds the cost of false positives (declining good applicants).

5. Model Performance

Metric	Value	Benchmark
AUC-ROC	0.6742	Acceptable > 0.60 for single-bureau scorecard
Gini Coefficient	0.3484	Acceptable > 0.40; below threshold -- see limitations
KS Statistic	See analysis.ipynb	Typically > 0.30 for production models
Calibration	Well-calibrated (LR)	Required for IRB PD estimation

Note: the Gini coefficient of 0.35 falls below the commonly cited 0.40 threshold for acceptable single-bureau scorecards. This is attributable to the limited size and age of the training dataset rather than a fundamental modelling deficiency. The discriminatory power is expected to improve materially with a larger, more recent dataset.

6. Known Limitations and Model Risk

Dataset size and vintage: 1,000 observations from 1990s Germany. Insufficient for a production IRB model under EBA minimum data requirements. Through-the-cycle representativeness is not established.

Income proxy: Income is estimated from installment rate and employment duration rather than observed directly. This introduces measurement error into the DTI ratio feature.

No macroeconomic features: The model produces through-the-cycle PD estimates and does not incorporate point-in-time macroeconomic adjustments (unemployment, GDP, interest rates). This limits its use for stress testing without overlay.

No population stability monitoring: Population Stability Index (PSI) monitoring is not implemented. Score distribution shifts would not be automatically detected.

No independent validation: The model has not been subjected to independent model validation as required under EBA/GL/2017/16 and the ECB Guide to Internal Models.

Fairness: Disparate impact analysis across protected characteristics has not been conducted.

7. Planned Enhancements

- Migration to Home Credit Default Risk dataset (307,511 applications, 2007-2011)
- Implementation of Population Stability Index (PSI) monitoring
- Stress testing module: PD uplift under adverse macroeconomic scenarios
- Formal backtesting against observed default rates by rating grade (Kupiec test)
- Disparate impact analysis across age and gender

References

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